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Leader-follower Motion Control of Two Line Following AGVs using PID Algorithm

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***Abstract*—The aim of this paper is to study the leader-follower motion problem of two line following AGVs that follow the same path and maintain the desired relative distance *l*. In this paper, through converting the relative angle ψ in the *l*-ψ model into the front or back position relationship *p* between two AGVs, a *l*-*p* control model is proposed. Based on this model, the tracking error relationship between the follower and the leader is established and the linear velocity of the follower is generated using the PID algorithm. The proposed method is tested on the self-developed experimental platform based on STM32, and the results show that the AGV can follow the specified path and maintain the desired distance.**

Keywords—line following AGV; leader-follower approach; PID algorithm; STM32

I. INTRODUCTION

AGV (Automated guided vehicle) refers to an intelligent vehicle equipped with an optical or electromagnetic guidance detection device that can travel along a prescribed guide path, and is a branch of a wheeled mobile robot [1]. Up to now, AGV is one of the main automated transportation systems used in practical applications [2]. In actual production, cargo in a factory or warehouse often have different weights and sizes, and some even exceed the transport capacity of a single robot. If AGVs of different sizes are customized for each cargo type, it will undoubtedly increase production costs and reduce the flexibility of AGV's using. Therefore, in this case, using multiple relatively small AGVs for cooperative transportation is a more optimal solution [3].

The cooperative transport of large cargo using multi-AGVs can be considered as a multiple mobile robot formation problem, since they need to maintain a certain formation. [4] designed two fuzzy controllers to control the distance and angle between two robots and used the depth camera of the follower robot to measure the distance and direction of the leader robot. [5] proposed a fuzzy compensated PID controllers based on the view of the leader-follower approach for formation control to improve the effectiveness of PID controller in nonlinear systems. [6] proposed a leader-follower formation control scheme with two independent fuzzy logic controllers for moving to the target and collision avoidance respectively, and used a simple transformation to fuse the two logics. [7] extends the trajectory tracking control of a single non-holonomic mobile robot to the formation control of two robots based on PI control techniques, where one is the leader and one is the follower. However, most of the studies have been conducted with AMR (Auto mobile robot) as the model, and there is little research on line following AGV that are equipped with multiple infrared sensors or magnetic sensors to sense the laid black or magnetic path and can adjust their angular velocity according to the degree of deviation from the path so that they do not go off the path.

In summary, this paper combines the characteristics of line following AGV, improves the *l*-ψ model of the traditional leader-follower approach, converts the relative angle ψ into the relative position *p*, and proposes the *l*-*p* control model. Based on this model, the tracking error relationship between the leader and the follower is established, and the PID algorithm is used to generate the linear speed of the follower to maintain the relative distance between them. Finally, according to the resources required for the approach, this paper builds an experimental environment for testing and achieved the expected experimental goal.

II. LEADER-FOLLOWER MOTION CONTROL

A. Line Following AGV Kinetic Equations

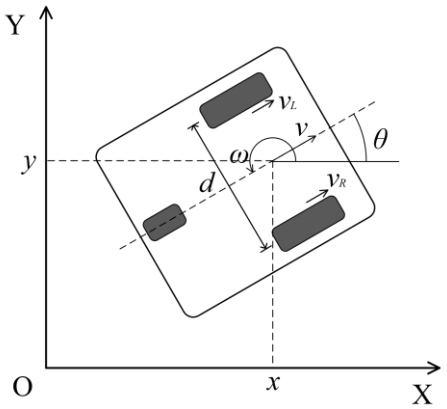


Fig. 1. Illustration of AGV

In this paper, we study the line-following AGV in which is a two-wheel differential drive robot model. As shown in Fig.1, (x, y, θ) is the pose of AGV in the global

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coordinate system. The following kinematic equation can be obtained:

$$\begin{cases} \dot{x} = v\cos\theta \\ \dot{y} = v\sin\theta \\ \dot{\theta} = \omega \end{cases} \quad (1)$$

Where v and ω is the linear velocity and the angular velocity of the AGV respectively. They can be calculated by:

$$\begin{cases} v = \dfrac{v_R + v_L}{2} \\ \omega = \dfrac{v_R - v_L}{d} \end{cases} \quad (2)$$

Where v_L and v_R are the speed of left and right drive wheels, d is distance between the two drive wheels.

B. An Improved Leader-Follower Approach based on PID Algorithm

Formation control of multiple robots refers to control the robots to reach the desired position in the team during the motion. [8] presents the classical leader-follower approach with l-ψ as its common controller, where the leader and the follower maintain the desired relative distance l and the relative angle ψ. The state variables of the system are $s = [l, \psi, \theta]^T$ and the control law of the follower $u = [v, \omega]^T$ is generated by standard I/O linearization techniques.

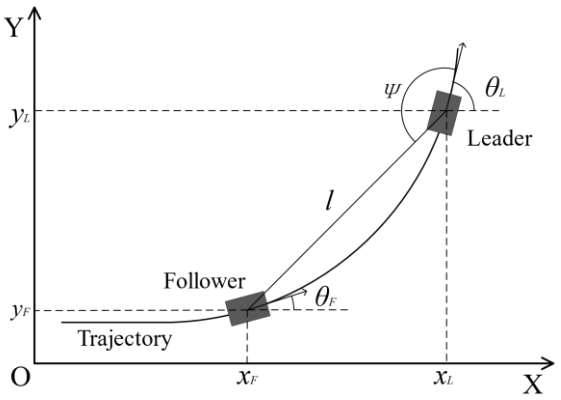


Fig. 2. Illustration of leader-follower model

In a multi-robot system, the l-ψ control model in which the leader and the follower need to maintain a fixed ψ. However, as shown in Fig. 2, the desired relative angle ψ between them varies when the AGVs are tracking on a curve path. Therefore, the l-ψ model does not work for controlling multiple line-following AGVs.

In addition, since the AGV tracks on the same path, the follower and the leader just to maintain the relative front or back position relationship p. Therefore, this paper makes an improvement to the classical leader-follower approach and proposes an l-p control model based on l-ψ control for solving line following AGVs' leader-follower motion. As described in the introduction, to ensure that the AGV does not deviate from a fixed path, the line-following algorithm can calculate the target angular velocity ω of the AGV at the next moment based on the current deviation from the path. Therefore, the l-p model only focuses on the target linear velocity v at the next moment. in summary, the state variables of the two AGVs formation control system improved by this paper are $s = [l, p]^T$ and the control variables are $u = [v]^T$.

Unlike l-ψ, the l-p model no longer focuses on the specific relative angle ψ, but translates it into the relative positions p of the leader and the follower. $p = 1$ when the follower is in front of leader, otherwise $p = -1$. Then, the tracking error relationship e_l between the leader and the follower is obtained based on p and l. Finally, the PID algorithm is used to control the error to obtain the target linear velocity v of the follower at the next moment.

As shown in Fig. 2, in the global coordinate system, let the pose of the leader be (x_L, y_L, θ_L) and the pose of the follower be (x_F, y_F, θ_F), then the relative distance l between them is:

$$l = \sqrt{(x_L - x_F)^2 + (y_L - y_F)^2} \quad (3)$$

The relative angles ψ is:

$$\psi = \pi + atan2\sqrt{\frac{y_L - y_F}{x_L - x_F}} - \theta_L \quad (4)$$

The value range of ψ is $(-\pi, \pi]$, which is then translated into the relative position relation p between the two AGVs as follows:

$$p = \begin{cases} 1, & \psi \in (-\pi/2, \pi/2] \\ -1, & \psi \in [-\pi/2, -\pi) \cup (\pi/2, \pi] \end{cases} \quad (5)$$

In order to unify the relationship between the error and the follower's linear velocity, i.e., to ensure that the follower accelerates when the error is greater than zero and decelerates when the error is less than zero. The following equation is used to calculate the follower error relationship e_l:

$$e_l = p(l_d - l) \quad (6)$$

Where: l_d is the expected relative distance between the leader and the follower, and l is the real distance. When the follower is before the leader, $e_l > 0$ if $l < l_d$; conversely, $e_l < 0$. When the follower is behind, $e_l > 0$ if $l > l_d$; conversely, $e_l < 0$.

Digital PID controllers are widely used in industrial control fields because of their simple control structure, easy parameter adjustment and no need to establish an accurate system model. In order to solve the problem that the integral term in the PID algorithm occupies large storage space and is difficult to calculate, this paper uses the incremental PID controller. The linear velocity increments are:

$$\Delta v(t) = k_p[e_l(t) - e_l(t-1)] + k_i e_l(t) + k_d[e_l(t) - 2e_l(t-1) + e_l(t-2)] \quad (7)$$

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The linear speed of the follower is:

$$v(t) = v(t-1) + \Delta v(t) \quad (8)$$

After generating the target linear velocity v and target angular velocity ω of the AGV by PID algorithm and line-following algorithm, the inverse solution is performed by equation (2) to obtain the velocity of the left and right drive wheels. Finally, the PWM output makes the left and right motors stabilize at that speed to achieve the purpose of maintaining the relative distance between the two AGVs. The flow chart is shown in Fig.3.

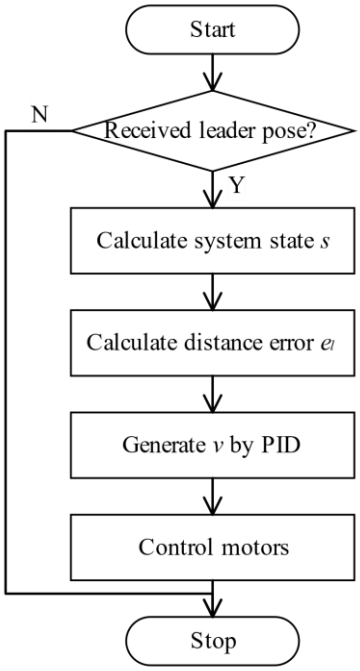


Fig. 3. Flow chart of control

III. EXPERIMENT

To shorten the product development cycle, most researchers use simulation to test their controllers. However, simulation is difficult to realize the perturbations that exist in the real world, resulting in simulation results that may be almost worthless. Therefore, in this paper, we conducted experiments on self-developed AGV platform based on the required resources for the controller.

A. Experimental platform

The AGV is a two-wheel differential drive mobile robot model with a universal wheel installed at its tail to maintain balance. The drive wheels have a diameter of 65mm and are driven by DC geared motors with high precision photoelectric encoders, with a distance of 160mm between the center points of the two drive wheels. The main controller uses STM32F407VET6 with main frequency up to 168MHz, and has FPU to accelerate the float point processing speed in the program. In addition, the MPU9250 can be used to observe the attitude of the AGV, and the ESP8266 Wi-Fi module completes the data transmission between the AGVs and between the AGVs and the PC. The hardware system architecture block diagram is shown in Fig.4.

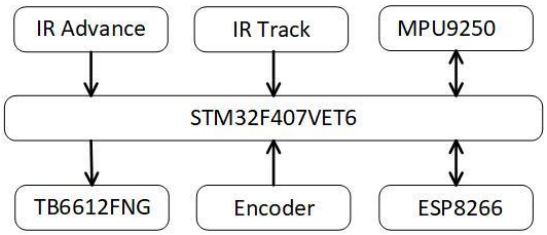


Fig. 4. Hardware system block diagram

There is no doubt that AGV is a complex control system and it is difficult to programming using bare-metal based structures such as polling and so on. In this paper, we developed a software system based on RT-Thread, an embedded real-time operating system, using its kernel functions to create and manage threads including position, motion control, communication and formation. The RT-Thread device framework allows separation of the underlying device driver from the upper layer application.

B. Data acquisition and calculation

As required by the approach, the leader needs to acquire its own pose data, then send it to the follower periodically. The follower reads its own pose data after receiving the pose data from the leader. The flow chart of pose data acquisition and calculation is shown in Fig.5.

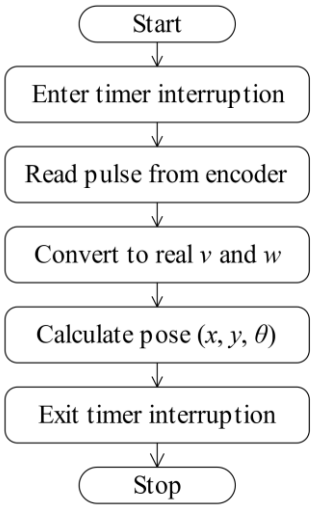


Fig. 5. Flow chart of pose data acquisition and calculation

C. Experimental design

The *l-p* control model proposed in this paper mainly addresses the application of the leader-follower approach to line following AGV, especially when traveling on a curve. Therefore, the path shown in Fig. 6 is designed in this paper. The path consists of three segments, which are 0.9m straight line, 0.8m radius length 1.25m curve and

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0.5m straight line, with a total length of 2.65m. Its starting coordinates are (0.0, 0.0) and ending coordinates are (1.7, 1.3).

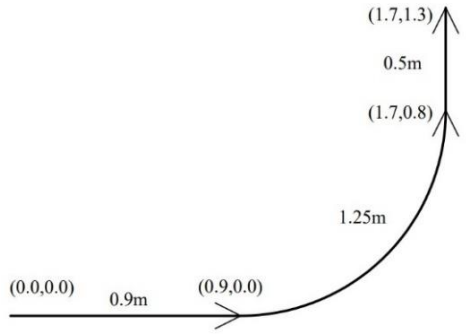


Fig. 6. Experimental path schematic

In the experiment, two AGVs follow the designed path above. The leader is in front and the follower is behind. Initially, the follower is at the starting point and the leader is at (0.5, 0.0). The leader tracks along the path at a constant linear velocity of 0.1m/s. The distance between the follower and the leader is maintained at 0.5m while following the path, i.e., l_d = 0.5m and p_d = -1. Referring to the experiments in the literature [4], the distance error is set to ±0.1m in this paper, i.e., the relative distance between the two AGVs should be maintained at 0.4m-0.6m during the motion. the linear speed of the follower is dynamically adjusted by the controller proposed in this paper.

D. Experimental results

The parameters set for the controller in this paper are k_p = 0.5, k_i = 0.0, and k_d = 0.05.

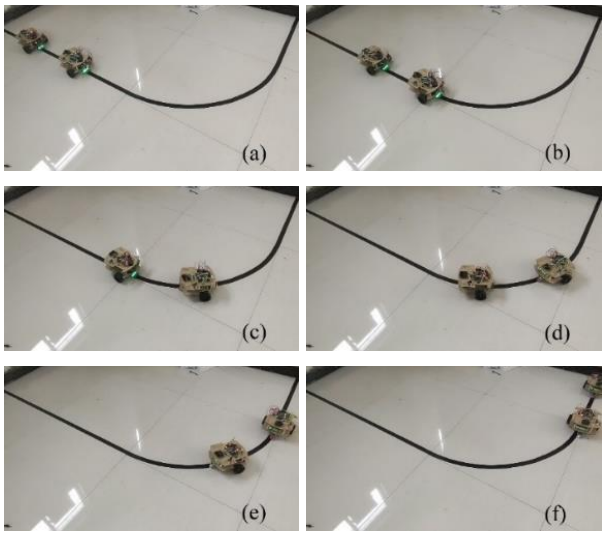


Fig. 7. Experimental snapshot

Some experimental snapshots are shown in Fig.7. (a) shows the starting point of the trajectory, which is in the stationary state; (b) shows the leader arriving at the beginning of the curve; (c) shows the follower arriving at the beginning of the curve; (d) shows the state when both are in the middle of the curve; (e) shows the state when the follower arrives at the end of the curve; (f) indicates that both are driving out of the curve.

The linear velocity between the leader and the follower is shown in Fig.8. The linear velocity of the leader is stabilized at 0.1m/s, and the linear velocity of the follower varies according to the distance between the two AGVs and is also gradually stabilized at 0.1m/s.

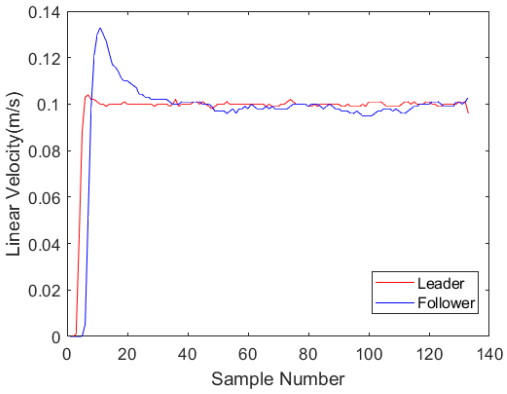


Fig. 8. Linear velocity curve

Fig.9 shows the relative distances between the two AGVs. The leader enters the curve at around sampling point 30 and drives out of the curve at sampling point 85; the follower enters the curve at sampling point 50 and drives out of the curve at sampling point 110. The maximum distance error between the two AGVs during the whole process is 0.059m, which does not exceed the maximum error limit of 0.1m.

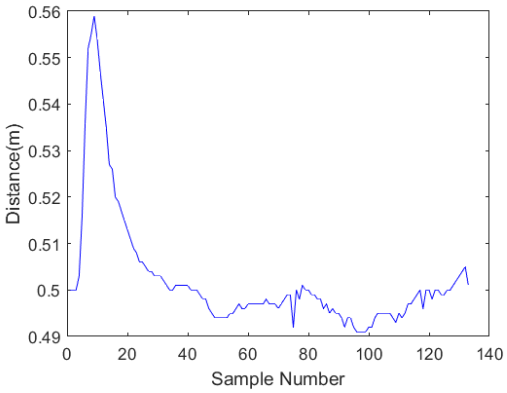


Fig. 9. Relative distance curve

IV. CONCLUSIONS

This work takes two line-following AGVs as the study object and solves their leader-follower motion problem. Firstly, this paper analyzes the reasons why the traditional leader-follower approach cannot be applied. Then, combining the characteristics of constant front-back position relationship of two-line following AGVs

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following on the same path, an *l-p* model is proposed, with *l* being the relative distance and *p* being the front or back position relationship. Based on this model, the tracking error relationship between the follower and the leader is established. Because the target angular velocity of the AGV can be controlled by the line-following algorithm, the PID algorithm is only used to control the error to generate the target linear velocity of the follower. Finally, a line-following AGV platform is constructed and experimented in this paper using STM32 microcontroller. The results show that the control method proposed in this paper can maintain the relative distance between two line-following AGVs.

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